12 Here are the first five terms of a number sequence.

45 40 35 30 25

(a) (i) Write down the next two terms of this sequence.

Subtract 5 each time: 20, 15.

1 mark
20, 15

..... , ...

(1)

A term of this sequence is −5

(ii) Which term?

45 − 5(n − 1) = −5; n − 1 = 10; n = 11.

1 mark
11th term

.....

(1)

The *n*th term of a different sequence is given by the expression $4n + 3$

(b) Find the 9th term of this sequence.

4 × 9 + 3 = 39.

1 mark
39

.....

(1)

(Total for Question 12 is 3 marks)

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